



Mishkat (مِشْكَاة) — the complete project file

Version of 6 October 2026. Author: Mohamed Nour Bou Ali. Site: <https://mishkatquran.org> (fallback: <https://mishk-at-4m1.pages.dev>). Code: <https://github.com/mednour87/Mishkat>. Every figure in this file comes from a file of the repository or from an official price page read on 6 October 2026; the source is given next to it.

1. Mishkat on one page

Mishkat is an AI-augmented Quran search engine, in Arabic and English, in which the AI never writes a religious word.

Type an idea, a question, a surah name, a verse number or part of a verse. Mishkat answers with:

- «الاجواب باختصار» **(the short answer)**: sentences **copied word for word** from the Quran text (Tanzil), *At-Tafsir Al-Muyassar*, *Al-Mukhtasar fi At-Tafsir* and HadeethEnc hadiths whose grade is known. Every sentence carries its reference;
- **the key verses with their complete tafsir**, then **the surahs ranked by relevance**;
- a **reader** verse by verse with Sheikh Mishary Alafasy's recitation **synchronised word by word**, inside a **3D galaxy of the Quran's 77,433 words**.

The name comes from the Verse of Light: ﴿مِثْلُ نُّورٍ كَمِشْكَاةٍ فِيهَا مِصْبَاحٌ﴾ (An-Nur 24:35). A niche (*mishkat*) gathers the light of the lamp; Mishkat gathers, around the visitor's question, the light of verified sources.

The project is offered to the whole Muslim community and will stay free: no ads, no account, no tracking (see §12).

2. The problem

1. **General chatbots invent verses.** Project measurement (`eval/human/auto_metrics.json` , 3 October 2026): 60 questions were put to a general chatbot (the same open model, gpt-oss-120b, without retrieval). Of the **121 quotations** it presented itself as Quran (inside ﴿﴾), **only 26 are exact**, **72 match no verse at all**, 17 are altered or merge two verses. Mishkat: **0 invented quotation, by construction**.
2. **Reliable tools are hard to use.** Concordances and tafsir libraries need the exact word, Arabic and patience. People who ask "what does the Quran say about...?" use chatbots.
3. **Sensitive questions** (legal rulings, people in distress, polemics, traps) need a caution that chatbots do not have.

3. The solution and its golden rule

Content	Comes from	Guarantee
Quran text	the Tanzil file (<code>public/data/core.json</code>), byte for byte (SHA-256 checked by the tests)	the model never writes a verse; no synthetic voice ever reads the Quran (human recitation only)
Tafsir, hadith, dhikr, statement of a ruling	copied word for word from the sources of the registry (<code>SOURCES.md</code>), with source and link	the AI only returns numbers taken from closed lists ; a number outside the list is dropped on the server and in the browser

Content	Comes from	Guarantee
Interface glue text	hand-written templates (Arabic, English)	no free text
Anything else	—	abstain and refer

For **rulings (حكم)** Mishkat never gives a fatwa: under a red «سؤال حساس» banner it shows the statement of **the Fiqh Encyclopedia of Ad-Durar As-Saniyyah** (dorar.net/feqhia, named by the challenge's scientific reference pack), word for word, with its link, and refers to alifta.gov.sa.

4. The site: the visitor's journey

1. **First visit:** write (or paste with one click) the basmala → choose one's place (country, city or position) → a **motion-graphics presentation film** on the Verse of Light recited by Sheikh Alafasy (skippable; shown once) → preferences (including age, for the child mode).
2. **Home:** one search bar (text or voice) above the 3D. **Since 6 October the shape shown at opening is وردة «السور» (the rose of surahs);** the visitor may choose another, and the choice is remembered.
3. **The answer:** short answer, key verses, ranked surahs; the camera flies to the verses in the 3D.
4. **Study:** the Mushaf and the tafsir side by side, the synchronised recitation, the recited word lit in the 3D.
5. **Daily tools:** in the ≡ drawer and the header bar (prayer, qibla, mosques, calendar, khatma, tekrar, adhkar, tasbih, statistics...).

5. Concept and design

- **Identity:** the starry night and the gold of the lamp. The **logo** was drawn after As-Sa'di's tafsir of 24:35: a glass mosque lamp that shines by itself "like a brilliant star" (كوكب دري), a lit wick inside, an olive sprig. The same drawing serves the logo, the icon and the animation (`public/img/logo.svg` → `tools/make_lamp.py` → `public/js/lamp.js`).
- **The 3D is the Quran itself:** 77,433 points = 77,433 words. **6 shapes** (galaxy, «قرآن», **rose of surahs**, dome, petals, challenge flower) × **7 orders** (Mushaf, revelation, Meccan then Medinan, number of words, of letters, of verses, average verse length). Each shape is **one continuous reading thread**: the camera follows the recited word without going back and forth (`public/js/readcam.js` , tested by `tests/reading.test.mjs`).
- **Typography:** Amiri Quran for the Quran, Amiri for religious texts, IBM Plex Sans Arabic and Inter for the interface, all self-hosted (SIL OFL licence).
- **Two themes:** dark (night) and light ("paper" panels around the night 3D). Contrast checked by `tools/ui_shots.mjs contrast` .
- **Mobile first:** checked at 375–390 px, RTL and LTR; exclusive panels (one open at a time), ≡ drawer on phones, ✕ full-screen button on every panel, `prefers-reduced-motion` respected, Wake Lock while reading.
- **Accessibility:** ARIA labels in Arabic and English, keyboard navigation (Escape closes the panel, V/O change shape and order), no essential text inside an image.
- **Installable app (PWA):** on phone and computer, from the browser; what was opened stays available offline (`public/sw.js`).

6. Code and architecture

- **Static site, no framework:** HTML, CSS and JavaScript ES modules (`public/`). Main files: `app.js` (orchestration), `engine.js` (search engine), `rag.js` (rules of the short answer), `galaxy.js` + `layouts.js` +

layout-worker.js (3D, three.js 0.160), practical.js (prayer, qibla, mosques), geomag.js (magnetic declination), hijri.js, khatma.js, tekrar.js, intro.js (film).

- **The engine runs in the browser**, in **Web Workers** (search-worker.js for search, layout-worker.js for the 3D shapes): the interface never freezes.
- **Serverless API**: Cloudflare Pages Functions (functions/api/*) — expand, select, answer, pick (AI), dense (question vector), tafsir, hadith, fatwa, mosques, tts, transcribe, health. **A single file calls a language model**: functions/_lib/selector.js (+ answer.js for the composer and the judge).
- **Data** built once by Python scripts (data_build/), frozen with a SHA-256 manifest.
- **Quality: 268 automated tests**, all green (npm test, Node, no npm dependency); evaluations in eval/; automatic phone/computer screenshots (tools/ui_shots.mjs); worldwide check of the practical tools (tools/check_world.mjs).
- **Traceability**: docs/TRACEABILITY.md lists every external host, every API and every data file; BASELINE.md (declared prior work) and CHANGELOG.md (work of the challenge window, **60 dated commits** since 4 October 09:00 Riyadh time); ARCHIVE.md (SHA-256 fingerprint of every file).

7. The search and RAG pipeline (evidence-bound extractive RAG)

Full description: mishkat/docs/MOTEUR_RECHERCHE_RAG.md and the plate docs/pipeline_rag.png.

#	Stage	AI?	Output
1	Question map : distress, child mode, off-topic, date/time, request for a tool (scope.js, tools.js)	no	fixed answer, tool opened, or next stage
2	Guard and routing : reference («2:255»), verification of a quotation, exact Quran words, safe spelling correction, prompt injection, personal fatwa, sensitive subject (engine.js, injection.js)	no	route
3	Understanding (/api/expand, gpt-oss-20b): intent + search keywords + proposed verses, never displayed , all verified	yes	search leads
4	Hybrid retrieval : lexical BM25 (verse, Muyassar, Mukhtasar, translation) + meaning (bge-m3, vector projected to 256 dimensions, ranked in the browser) + the AI's keywords + Quranpedia's human subject index (6,100 topics), rank fusion	vector only	closed list of at most 36 verses
5	Selection (/api/select, gpt-oss-120b): IDs from the list with a score (2 = answers, 1 = related)	yes (IDs)	confirmed verses
6	Short answer : numbered sentences (verse, tafsir, graded hadiths) → composer (120b) picks 1–2 IDs per concept → independent judge (20b, another prompt) keeps what answers → fixed rules R1–R13	yes (IDs)	1 to 3 points, each copied word for word
7	Complementary sources : Sunnah (HadeethEnc), ruling (Dorar Fiqh Encyclopedia), quoted hadith (Dorar), objections (Bayyinat)	selection only	cited cards with link

What is displayed is always **the browser's own copy** of the source text, never text returned by a model. A concept without evidence is declared "not covered". **Without AI** (quota exhausted, outage) the deterministic engine still answers, always sourced, marked "not confirmed by AI".

Measurements (results files):

- RAG on **1,000 questions** Arabic/English, independent judge of another model family (DeepSeek-V3.2): first verse answers directly **84.5 % [81.9–87.1]**; short answer on topic **99.5 %**; **0 / 1,058** passages not found word for word in their source; critical cases **15 → 2** (eval/rag1000/README.md).

- Public benchmark **Qur'an QA 2023, Task A** (51 scored questions, official scorer, 3 runs on the free tier): **MRR@10 0.616 [0.490–0.742], MAP@10 0.300 [0.210–0.402]**. The best published system (0.576 / 0.313, arXiv 2412.11431) lies inside our intervals: **comparable, not better** (eval/qqa23/ci_free3.json).
- Map of 1,000 questions without AI: **630 / 671** expectations met; **50 / 50** legitimate questions "that look sensitive" not refused.

8. The features, one by one

1. **Search and short answer** — see §7. Typed, spoken (ElevenLabs Scribe, then Whisper) or **by reciting a verse** to find it.
2. **Quotation check** — "is this a verse?": exact / altered word highlighted with the exact verse / two verses merged / not in the Quran; if it is not a verse, the hadith is looked up in Dorar with the muhaddith's verdict.
3. **Rulings (حكم)** — red «سؤال حساس» banner, the Dorar Fiqh Encyclopedia's statement word for word, consensus or difference as it states them, referral to alifta.gov.sa. Never a fatwa.
4. **Stories of the prophets** — «علوم السور» of the Al-Jamhara encyclopedia and the Quranpedia index; verses from the Tanzil text.
5. **A person in distress** — a fixed support message and findahelpline.com first, then verses of hope with their full tafsir, never a verse of punishment.
6. **Traps and injections** ("ignore your instructions", "write me a hadith") — fixed answer, **no AI call**; the API refuses these texts too.
7. **Off-topic** (recipe, price, weather, code, homework...) — a fixed, kind answer that says what Mishkat does; never sent to a model.
8. **Date and time** — immediate answer from the device's clock: Gregorian date, Umm al-Qura Hijri date, local time.
9. **Child mode** (under 18) — no fatwa, no red; a gentle invitation to memorise a surah or follow a khatma.
10. **Reader** — the Mushaf verse by verse; **Alafasy's recitation synchronised word by word; 8 tafsirs**: Al-Muyassar, Al-Mukhtasar (ar/en), As-Sa'di, At-Tabari, Ibn Kathir (Arabic, and abridged in English), Al-Baghawi, Al-Qurtubi; English translation; transliteration; tajweed colours (optional) and a tajweed rules page; «about the surah».
11. **3D galaxy** — 6 shapes × 7 orders, **rose of surahs by default**; resting 3 s on a star shows its word, verse and frequency; surahs read light up.
12. **Khatma v2** — «اختر لي / choose for me» (minutes a day, moments after each prayer, pace, short surahs first, deadline); plan over the 604 pages; automatic catch-up; "I finished this surah ✓"; reading record; .ics reminders; the 3D lamp lights up with the surahs read.
13. **Tekrar (memorising)** — number of verses and repetitions, listening to the sheikh, a counter that counts a repetition only after 70 % of the reciter's time; an encouraging mark from 7/10; an optional **memorisation test** (write or recite, word-by-word check, no generated text); sharing and inviting.
14. **Adhkar** — HadeethEnc and Hisn al-Muslim, **each dhikr kept only if Dorar gives it the grade صحيح or حسن**; **tasbeeh** counter.
15. **Prayer times** — Aladhan, method and authority shown, proposed by country; countdown; optional adhan or tone; month cached.
16. **Qibla** — compass with magnetic declination correction (§9).
17. **Nearby mosques** — OpenStreetMap list and a Google Maps map in the page, route.
18. **Hijri calendar** — today's date, converter both ways, **reviewed** remarkable days, each with its verses and authentic hadiths; disputed commemorations are not listed.

19. **Statistics** of a surah and a word (counts only, no numerology); monthly **progress**; **3D engagement map** (three levels named after 24:35).
20. **Useful links** (sources of the challenge's reference pack), **settings**, **export / restore / erase** of the visitor's data (it stays in their browser).
21. **Voice search** — ElevenLabs Scribe v2 with Mishkat's vocabulary (the 114 surah names and common terms), Whisper as fallback: **30/30** on 30 spoken questions in ten accents (`eval/stt/RESULTS.md`).
22. **Verified answers without AI** — counts (verses, surahs, juz, sajdahs...) **computed** from the site's Tanzil data; basic questions (pillars of Islam and faith, the greatest verse, the first revelation...) answered by an **authentic hadith** from HadeethEnc with its text, grade and link (`public/js/facts.js` , `tests/facts.test.mjs`).
23. **Approved encyclopedias of the challenge pack** — creed (dorar.net/aqeeda) and history (dorar.net/history) for questions no verse answers directly, quoted word for word with their link.
24. **Practical fiqh** («مفصلات الصيام» , «نواقض الوضوء» , «كيف أتوضأ» , «شروط الصلاة») — an index of **2,095** headings of the worship books of the Fiqh Encyclopedia (dorar.net/feqhia): the section whose heading names the subject is shown with its headings as written, each linked to its text and evidence; no matching section, nothing is shown (`public/js/fiqhpractical.js`).
25. **The Most Beautiful Names** (narration of at-Tirmidhi) — list, full listening with each name timed, learning group by group; a **tasbeeh counter**; the **light of the month**: each completed act of worship sends a ray to the lamp.
26. **Pure full-screen view** — only the stars and the recited word, with the logo and the name «مشكاة» and a clear way back; a ray leaves the last word of a verse when it ends; a bar of light turns slowly (one turn a minute) and lights only the stars it crosses, in their own colour.
27. **Reminders on phone and computer** — prayer times (on time or before), remarkable days, the white days, Monday and Thursday fasting, khatma reading moments, as device notifications with sound and vibration; plus an .ics calendar file that rings while the site is closed (`public/js/notify.js`).

9. Worldwide validity: qibla, clock, date, prayer times, mosques

Check of 6 October 2026 on **33 places on every continent**, including Svalbard (78° N), Iqaluit, Ushuaia, Apia (UTC+13) and Kiritimati (UTC+14) — `tools/check_world.mjs` , result `eval/world/check_world_2026-10-06.json` , tests `tests/world_oct6.test.mjs` and `tests/geomag.test.mjs` .

- **Qibla** — initial **great-circle** bearing from the place to the Kaaba (21.4225, 39.8262), the standard spherical-trigonometry formula, computed in the browser. **Compared with Aladhan's qibla API (an independent computation): 0.000° difference at all 33 places.**
 - **Fix of 6 October:** a phone compass shows **magnetic north** (Android: absolute orientation; iOS: `webkitCompassHeading`). The qibla is computed from **true north**. The difference (declination) ranges from a few degrees in the Arab countries to **20° in New Zealand and 24° in northern Canada**. The dial now adds the place's declination computed with the **World Magnetic Model WMM2025** (NOAA and the British Geological Survey, official coefficients copied as they are), **checked against the 100 official test values** shipped with the model. It also shows the angle to read on a magnetic compass. Near the magnetic pole (the WMM "blackout zone", horizontal field < 2,000 nT) it does not turn and says why. At the Kaaba itself: «استقبل الكعبة بالمعانة» (face the Kaaba by sight).
- **Prayer times** — Aladhan with the place's coordinates: the time zone is the place's (returned by Aladhan), not the visitor's. **Fixes of 6 October:** the month loaded is **the place's** month (a city across the date line could show the 1st of the month as "today"); at high latitudes Aladhan's "angle based" rule is explicit and explained above 48°; the country of a GPS position (hence the proposed method) is the country of the **nearest**

GeoNames city (before, only 23 time zones were known and the rest of the world got Tunisia's method). At all 33 places the place's day is found and the six times are in order.

- **Clock and date** — the device's clock (the visitor's local time); Hijri date from the browser's **Umm al-Qura** calendar (`Intl` , ± 2 days adjustable to follow local moon sighting). **Fix of 6 October**: a browser without the Umm al-Qura calendar showed the Gregorian year as the Hijri year; it now uses the **tabular** Islamic calendar (the classical rule of 11 leap years in 30), which may differ by one or two days.
- **Mosques** — OpenStreetMap via Overpass (several mirrors tried two at a time), anywhere in the world: checked in London, Toronto, Auckland, Tokyo, Reykjavik, São Paulo, Jakarta (80 mosques). Overpass's public servers can be saturated for a few seconds (Jakarta failed once, then answered); the page's Google Maps map always works. Coverage depends on what OpenStreetMap contributors have mapped.

10. External services and sources

Religious sources (full registry: `mishkat/SOURCES.md` , S1–S34): Tanzil (Quran text, CC BY 3.0); QuranEnc (Al-Muyassar, Al-Mukhtasar, Noor International translation); Quran.com (As-Sa'di, Ibn Kathir, Al-Baghawi, Al-Qurtubi, Alafasy's recitation and word timings, transliteration); Quranpedia (At-Tabari, subject index, surah information); HadeethEnc (translated, graded hadiths); Dorar (hadith encyclopedia, fiqh encyclopedia); Al-Jamhara (surah sciences); Hisn al-Muslim; Bayyinat (titles and links only); `cpfair/quran-tajweed` (CC BY 4.0).

Technical services: Cloudflare Pages + Functions (hosting, API); Cloudflare Workers AI (`bge-m3`); Groq (`gpt-oss-120b`, `gpt-oss-20b`, Whisper, Arabic Orpheus); ElevenLabs Scribe (voice search transcription); OpenRouter (capped paid backup); Aladhan (prayer times); OpenStreetMap / Overpass / Nominatim (mosques, city search); Google Maps (embedded map, keyless); GeoNames (cities, CC BY 4.0); NOAA/BGS WMM2025 (magnetic declination, public domain); Wikimedia Commons (adhan recording, CC BY-SA 4.0); ElevenLabs and Microsoft Edge read-aloud voices (narration of the video and film, **never a verse**). Tools declared under clause 9/15 of the rules (`TOOLS.md`).

11. Security and privacy

- **Headers** (`public/_headers`): strict Content-Security-Policy (the site's own scripts + one hash, no remote script, no `unsafe-eval`), `frame-ancestors 'none'` , HSTS, `nosniff` , Referrer-Policy , Permissions-Policy (microphone and location only on the site, camera and payment forbidden), COOP.
- **API** (`functions/_lib/guard.js` , `handler.js`): **same-origin POST only** (another website cannot spend the AI quota); **per-IP rate limit** per route; maximum request size (64 KB, 4 MB for audio); **injection and off-topic texts are refused by the server** before any model; **daily ceiling** of paid AI calls (`DAILY_AI_CALLS`) and a credit limit on the key; circuit breakers per provider.
- **Secrets**: keys only in Cloudflare secrets and in `.dev.vars` (git-ignored); a test checks that no key is in the public files (`tests/security.test.mjs`).
- **Content**: tafsir HTML is reduced to text on the server; every displayed string is escaped.
- **Privacy**: **no account, no ads, no tracking**; questions are not stored; preferences, khatma and progress stay in the visitor's browser (export and erase available); the position is asked only after a click, never stored by Mishkat, and reaches the server only rounded to ~100 m for the mosque search; OpenRouter is called with `data_collection: deny` .

12. Cost per month

Today (launch traffic):

Item	Monthly cost	Source
Hosting + API (Cloudflare Pages, free plan: 100,000 Functions requests a day, 10 ms CPU per request)	\$0	developers.cloudflare.com/workers/platform/pricing
AI, normal case: Groq free tier first (per model: 1,000 requests/day, 200,000 tokens/day)	\$0	console.groq.com/docs/rate-limits
AI, backup: OpenRouter, pay per use	≈ \$0.0014 per question measured (<code>eval/cost_per_question.mjs</code>); public price of gpt-oss-120b on 6 Oct: from \$0.037 per million input tokens and \$0.17 output	OpenRouter API
bge-m3 vectors (Workers AI): 10,000 free "neurons" a day; bge-m3 = 1,075 neurons per million tokens	\$0	developers.cloudflare.com/workers-ai/platform/pricing
Domain mishkatquran.org	yearly .org fee (registrar invoice)	—
Narration (ElevenLabs)	used once, files shipped with the site: the subscription can be stopped with no effect on the site	elevenlabs.io/pricing

Running total: ≈ \$0 a month + the domain + a few dollars of paid backup depending on traffic.

13. Going live for many Muslims: what to increase

Estimate from the measured cost (\$0.0014 per question if the whole question goes through the paid backup; about 5 API calls per question):

Questions a day	AI (if all paid)	Cloudflare	Order of magnitude
1,000	≈ \$42/month	\$5 (Workers Paid advised)	≈ \$47/month
10,000	≈ \$420/month	\$5 (1.5 M requests, under the 10 M included)	≈ \$425/month
100,000	≈ \$4,200/month without cache	≈ \$6.5 + CPU	to be cut sharply by the shared cache

These are ceilings: frequent questions are already pre-computed and identical requests cached.

To increase, in order:

1. **Cloudflare Workers Paid** (\$5/month: 10 M requests and 30 M CPU ms included); **global rate limit and daily ceiling** in KV or Durable Objects (today in the memory of each instance).

2. **Shared cache of AI answers (KV)**: a popular question is paid only once.
3. **AI contract**: Groq's paid tier or OpenRouter credit with alerts; keep the no-AI engine as the safety net.
4. **Public dependencies**: move to the Quran Foundation's authenticated API; a dedicated Overpass/Nominatim instance or a cache (Nominatim's policy allows 1 request per second); local prayer-time computation as a backup to Aladhan.
5. **Monitoring**: Cloudflare dashboards and alerts, error reporting.
6. **Scholarly board**: human review of reported cases, blind human rating and a user test (kit ready in `eval/human/`).
7. **More languages** from the QuranEnc catalogue, more reciters.

14. Alignment with the challenge rules (Islamic AI Challenge 2026)

Requirement	Where
Track 01 «الحوار المعرفي والإجابات الموثوقة» (knowledge dialogue and reliable answers)	the whole engine; the practical tools are called from the conversation
A working product, not a prototype	live site, 268 green tests, worldwide check
Live demo	https://mishkatquran.org (fallback mishkat-4m1.pages.dev)
Public GitHub repository	github.com/mednour87/Mishkat (to be made public on 6/10)
Video ≤ 2 min	04_LIVRABLES/video/ (119.9 s)
PDF/PPTX presentation	04_LIVRABLES/Mishkat_presentation.pdf and the Arabic problem–solution–mechanism deck
Registry of sources, licences, tools	SOURCES.md , NOTICE.md , TOOLS.md , docs/TRACEABILITY.md
Prior work declared; only the window's work is judged	BASELINE.md , tag baseline-2026-10-03 , CHANGELOG.md (dated commits T090–T116)
Synthetic or anonymised data	no user data; synthetic or public benchmarks
Sources of the scientific reference pack	rulings: Dorar feqhia; stories: Al-Jamhara; binbaz.org.sa removed on 4/10
External tools declared (clause 9/15)	TOOLS.md
Intellectual property (clause 13)	LICENSE "All Rights Reserved": clause 13/7 forbids an open licence without written agreement; the organisers keep their evaluation licence
Final grid (tech + AI 25, measured benefit 20, reliability 15, innovation 15, UX 10, operational realism 10, clarity 5)	§7 (AI), §2 and §7 (measurements), §3 (reliability), §5 and §8 (3D, recitation), §11–13 (operations)

15. A gift to the community

Mishkat is offered to the whole Muslim community, **free and for good**: no ads, no account, no resale of data. The architecture makes it possible: the base cost is almost zero and every answer stays correct even without AI. "All rights reserved" protects the work during the challenge (clause 13/7); it does not prevent free access. After the challenge the author may choose an open licence with the organisers' written agreement, and seek support (waqf, patrons, institutions) to carry the load if traffic grows.

﴿...نُورٌ عَلَى نُورٍ يَهْدِي اللَّهُ لِنُورِهِ مَن يَشَاءُ...﴾ (An-Nur 24:35)

